# CSE 400: Fundamentals of Probability in Computing

Project Scribe Report (Milestone 1)

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**Topic:** Stochastic Optimization of Spatial Throughput

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**Project Summary**

Our project focuses on Wireless Networks. Specifically, we are using probability to find the perfect balance between **Reliability** (messages getting through clearly) and **Throughput** (total data speed) by using a smart scheduling algorithm called DPP (Determinantal Point Process).

## Scribe Question 1: Project System and Objective

*What is the probabilistic problem being addressed in your project? Clearly state the system objective and identify the primary sources of uncertainty involved.*

### Answer:

#### 1. The Core Problem: Speed vs. Reliability

The main problem we are solving is to find the best optimization in trade-off between **how many** devices can talk at once versus **how clear** their output is.

Current research focuses too much on **"Reliability"** (ensuring zero interruptions). This results in a network that is safe but very slow because it forces devices to wait too long to speak. Our project addresses the uncertainty of finding the **"Sweet Spot"** the optimal density of devices that maximizes the total speed of the network without causing a crash.

#### 2. The "Classroom" Analogy (Simplifying the Logic)

To understand the problem we have take this simple example to make it understood better to non technical readers, imagine a large classroom full of students:

- **The Old Way (Aloha Protocol):** The teacher asks a question, and the answers by students are randomized in this approach it's like flipping a coin and if you get heads then you answer otherwise you don't. This method is too random and uncertain and hence inefficient, The room is loud, chaotic, and the teacher understands nothing. (High Interference, Zero Success).

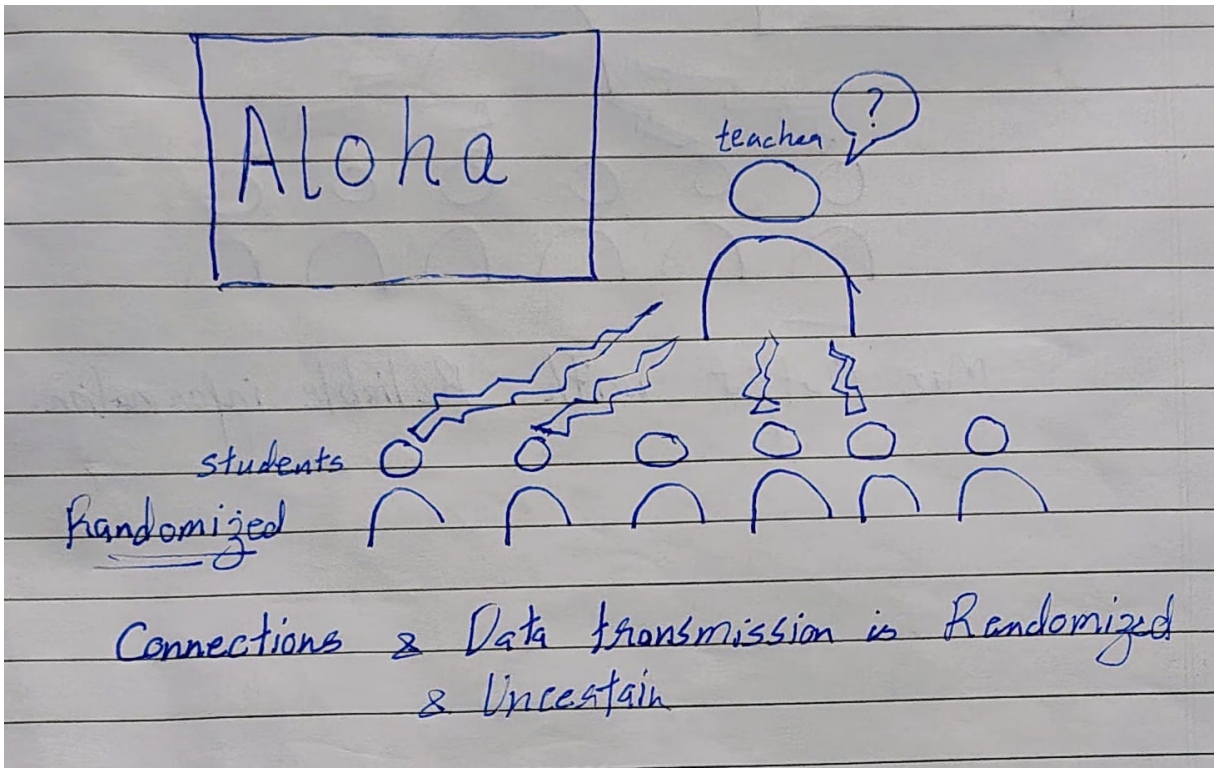


Figure 1: Aloha Simplified

- **The Safe Way (DPP Approach used in paper):** The teacher strictly enforces silence. Only one student in group is allowed to speak whereas it neighbors are completely shut off.This way we get best reliability and the answers are very clear, but the class is moving very slowly because very less people are participating. (High Reliability, Low Speed).

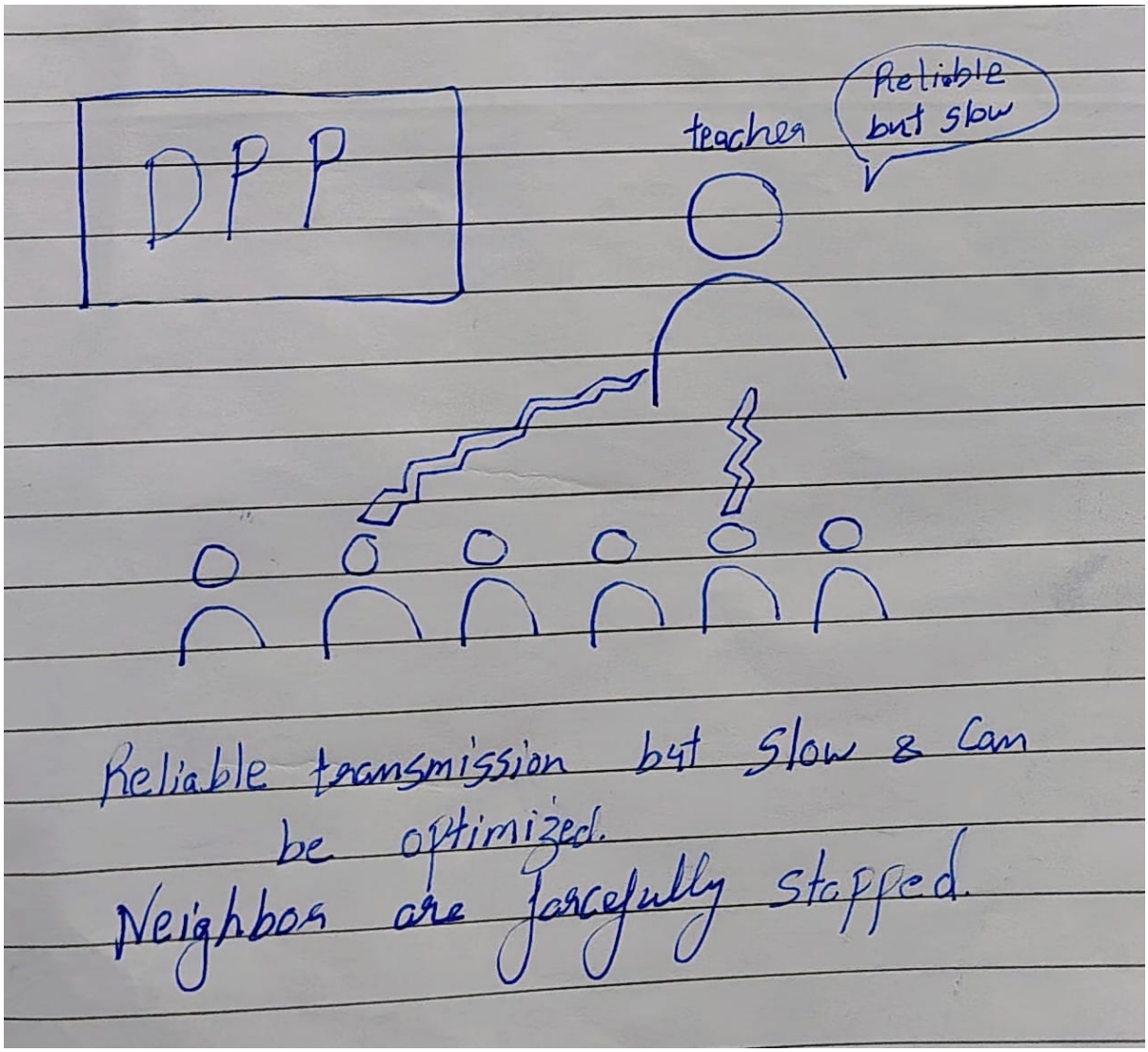


Figure 2: DPP Method

- **Our Solution (The Optimization):** We use probability to calculate the perfect seating arrangement. We want just enough distance between students so that many can talk simultaneously without drowning each other out. We are trying to find the specific number of students (Density) that creates the best reliability as well as perfect and clear answers (Data) of information.

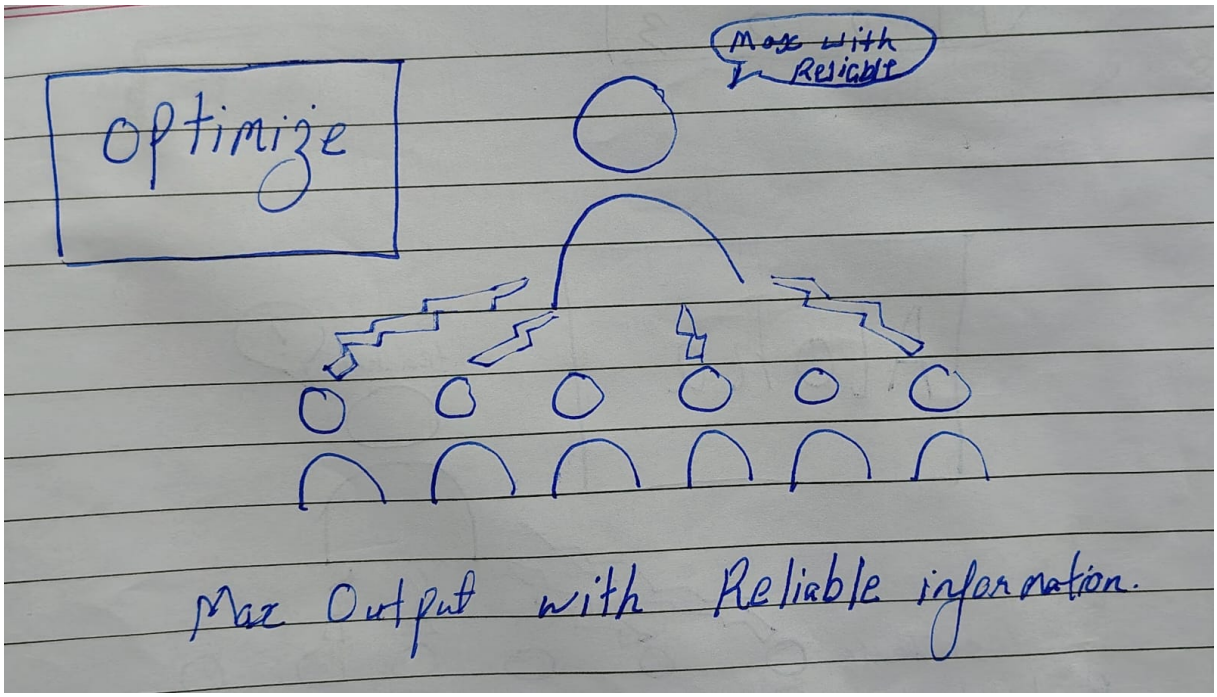


Figure 3: Optimized Method

### 3. System Objective

Our mathematical goal is to maximize the **Spatial Throughput Function**:

$$\text{Total Speed} = (\text{Number of Talkers}) \times (\text{Probability of Success})$$

If we add more talkers, the "Probability of Success" goes down. We need to find the exact point where this multiplication gives the highest result.

### 4. Primary Sources of Uncertainty

- **Location Uncertainty:** We don't know where users are sitting. They are scattered randomly (like raindrops).
- **Signal Fading:** Signals fluctuate randomly. Sometimes a signal is strong, sometimes it is blocked by a wall.
- **Interference:** The noise level depends on how many **other** people decide to talk at the exact same time, which is unpredictable.

# Scribe Question 2: Key Random Variables and Uncertainty Modeling

*Identify the key random variables in your project and describe how uncertainty is modeled for each. Clearly state any probabilistic assumptions currently being made.*

## Answer:

### 1. Key Random Variables

Variable Definition Table

| Variable | Simple Explanation & Model |
|---|---|
| **Device Locations** ($\Phi$) | **Where people are sitting.** We assume people are scattered completely randomly. *Model: Poisson Point Process (PPP).* |
| **Active Talkers** ($\Phi_{DPP}$) | **Who is allowed to speak.** This is a subset of people chosen by our algorithm to ensure social distancing. *Model: Determinantal Point Process (DPP).* |
| **Signal Strength** ($H$) | **The volume of the voice.** Sound does not travel perfectly; it fades and fluctuates. *Model: Exponential Distribution (Rayleigh Fading).* |
| **Interference** ($I$) | **The background noise.** The sum of noise coming from everyone else in the room. *Model: Stochastic Sum of Random Signals.* |

### 2. How We Model Uncertainty

- **Assuming Randomness (Poisson Process):** We assume the network starts in the "Worst Case" scenario where devices are uncoordinated and scattered randomly. This ensures our solution works in messy, real-world environments, not just in a lab and also it is scalable in nature.

- **Assuming Fading (Rayleigh Model):** We assume that signals bounce off walls and obstacles. In probability terms, this means the signal strength is an **Exponential Random Variable**. It is memory less knowing the signal was strong a second ago doesn't guarantee it will be strong now.

- **Modeling Repulsion (The Kernel):** This is the core of our project. We model "Repulsion" (Social Distancing) using a matrix. If two devices are too close, the probability of them both being active drops to zero. This is a mathematical way of forcing devices to be in coordination with each other so that their signals don't collide and mess up the transmission signal sent.

# Scribe Question 3: Probabilistic Reasoning

*Explain how probabilistic relationships (e.g., dependence, independence, conditionality) are used within your project system to support reasoning, inference, or decision-making.*

## Answer:

### 1. "Negative Dependence" (The Politeness Rule)

The most important concept in our project is **Negative Dependence**.

- In a normal network (Independence), if User A talks, it has no effect on User B. They might collide.
- In our project (Dependence), the system is designed so that if User A talks, the probability of User B talking *decreases*. They are "Negatively Dependent." This probabilistic rule automatically prevents collisions without needing a central boss to manage everyone.

### 2. Conditional Probability (The "If-Then" Logic)

We use conditional probability to calculate success. We ask:

> *"**Given** that User A is talking, what is the probability that the noise from others is low enough for him to be heard?"*

Because of the "Politeness Rule" mentioned above, the condition "User A is talking" implies "Neighbors are silent," which boosts the probability of success.

### 3. The Optimization Loop

We use a simple reasoning loop to find the best answer:

1. **Guess:** Pick a density (e.g., "Let's try 50 people in the room").
2. **Simulate:** Use the Probability formulas to simulate the noise levels.
3. **Measure:** Calculate the total successful data speed.
4. **Reason:** If adding more people increased the total speed, we guess a higher number next time. If the speed crashed, we guess a lower number.

# Scribe Question 4: Model-Implementation Alignment

*Describe how the current probabilistic model aligns with the project's implementation or experimental setup. Highlight any assumptions that influence design or evaluation choices.*

## Answer:

### 1. The "Seating Classroom" Analogy

The Python implementation which is given in the research paper github aligns with the probabilistic model by acting like a photograph rather than a movie,it showcases that we take a snapshot from that time frame instead of looking it as whole.

Let me again take a classroom example, Imagine trying to optimize a classroom. You wouldn't need to film the entire semester. You would just need to take a **"Seating Chart Photo"** (a Snapshot) of one specific moment to see if students are sitting too close to each other. This is better than monitoring the devices at each time frame.

- **The Probabilistic Model:** Assumes that at any given instant, devices are distributed according to a "Spatial Snapshot" (Poisson Point Process).
- **The Python Code:** Mirrors this exactly. It doesn't simulate time passing or packets moving. Instead, it generates one static map of devices, checks who can hear whom, and calculates the success probability for that single instant.

### 2. Why this Alignment Matters

This "Snapshot" approach aligns perfectly with our objective of finding **Spatial Throughput**. We don't care about when a packet arrives (Time); we care about where the devices are located (Space).

- The code assumes **Rayleigh Fading** (Exponential Random Variables) for signal strength.
- The code assumes **Static Nodes**, meaning users don't run around the room while talking.

These assumptions allow us to run thousands of "Snapshot Simulations" quickly to find the average performance, rather than waiting for a complex time-based simulation to finish.

# Scribe Question 5: Cross-Milestone Consistency

*Describe the current state of your project's probabilistic model. Identify the assumptions that are presently well-defined, and indicate which aspects are expected to evolve.*

## Answer:

### 1. What is Defined (The "Physics Engine")

At Milestone 1, we have successfully defined the "Physics" of our world. We know exactly how the environment behaves:

- **The Environment:** We have locked in the **Rayleigh Fading Model**. We know that signals will fluctuate based on an Exponential Distribution.
- **The Scheduling Logic:** We have defined the **Determinantal Kernel**. This is the math that enforces "Social Distancing" between devices.

### 2. What Will Evolve (The "Throughput")

While the physics are set, the **Optimization Strategy** is what we are building next. Currently, the code calculates "Fairness." In the next milestone, we will evolve it to calculate **"Speed"**.

We plan to run a specific experiment called **"The best path"**:

- **Step 1:** We will run the simulation with a density of $\lambda = 10$ (10 users). The network will be reliable but slow.
- **Step 2:** We will increase to $\lambda = 50$. The speed will increase.
- **Step 3:** We will increase to $\lambda = 100$. The interference might become too loud, and the speed will crash.

We expect the model to evolve from a simple calculator into an Optimizer that can automatically find the peak of this "Throughput Hill."

# Scribe Question 6: Open Issues and Responsibilities

*Identify any unresolved probabilistic questions or ambiguities in the project at this stage, and indicate which role(s) are responsible for addressing them.*

## Answer:

### 1. The Big Open Questions

There are three major questions we still need to answer:

- **The Tipping Point:** At exactly what density does the network crash? We know it happens, but we haven't found the number.
- **Sensitivity:** If the environment changes (e.g., more walls/obstacles), does our "Optimal Density" change drastically? Or is our solution robust?
- **Gain Qualification:** We know our method is better than random guessing (Aloha), but how much better? Is it 10% faster or 200% faster?

### 2. Assigning Responsibilities

To solve these, we have divided the work for the next milestone:

- Some of us will research and put our analysis together and change the codebase accordingly to get the results and interpret in our analysis to calculate the throughput.
- Other will use matlab to generate models which will help us to scale and look at the whole picture clearly.